STARSHIP AS AN ENABLER FOR FASTER, MORE AMBITIOUS MISSIONS TO EUROPA

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Abstract. Europa's astrobiological promise is matched by the severity of its engineering constraints. The propulsive energy required to reach the surface of Europa diminishes potential science payloads to the point at which historical flagship mission plans, using expendable rockets like the SLS, would deliver less than 50kg to Europa, with a transit time of over 6 years. In this paper, we show the feasibility of the SpaceX Starship delivering payloads of 10000kg or more, directly to the surface of Europa, with transit times of less than 2.5 years.

Introduction

Europa, the icy moon of Jupiter, is a prime target in the search for extraterrestrial life[1]. It is thought to have a deep, salty ocean[2] under a layer of ice, possibly with conditions similar to those that are thought to have led to the emergence of life on Earth[3]. There has, however, been no mission to the surface, nor is one under development by NASA. Historical planning for missions to Europa has been heavily constrained by the low payload mass and lengthy transit times required for spacecraft to reach Europa. The 2016 Europa Lander proposal, as analyzed in the 2023-2032 decadal survey[4][5] called for launch on the SLS Block 1B in the mid 2020s. The high delta V (DV) requirements (approximately 11km/s between Earth orbit and Europa landing) allow 42.5kg of scientific payload to be brought to the surface of Europa. The lander was expected to operate for 20 days; the low mass precluded radioisotope thermal nuclear generation (RTG) power and heavier shielding to protect electronics for a longer duration surface mission. These limiting factors contributed to the selection of different flagship priorities in said decadal survey. With the advent of reusable super heavy lift launch vehicles, like Starship, it is feasible to send much heavier payloads on much faster trajectories. Numerous new proposals take advantage of these capabilities[6][7][8][9] and we extend this analysis to Europa. We show that with near term technology, already under development for NASA's Artemis program, the scope of Europa exploration can be massively increased, with Starship potentially able to deliver 10T of payload to the surface of Europa, less than 2.5 years after launch from Earth. 10T of payload would dramatically ease engineering tradeoffs across a wide range of ambitious Europa exploration concepts. Cryobot design/mission design for true exploration of Europa's oceans requires science payload on the order of hundreds of kg to tonnes[10] [11]. We believe Starship is a prime candidate to deliver cryobots and other heavy scientific instruments to Europa.

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Mission plan and trajectory

Missions to Jupiter have followed a variety of trajectories, with transit times varying from 405d (New Horizons), 688d (Voyager 2) to a planned 3000d (JUICE), depending on whether the mission is/was a flyby, or captured into Jovian orbit (thus having to carry fuel to slow down at Jupiter). With the increased propulsive energy capabilities of Starship, we set 600d as an achievable Earth-Jupiter transit time leaving sufficient propellant to phase rapidly through the Jovian system enabling total time from launch to landing to be less than 2.5 yr. We analyze a baseline trajectory that uses 9.8km/s of DV (from the elliptical Earth orbit the vehicle departs from), taking just under 2.5 years to reach Europa, a fast trajectory taking 10.8 km/s in just over 2 years and a slow trajectory taking 4 years, using 8.8km/s of DV, while taking a lower radiation dose. Appendix A contains details on the launch profiles, gravity assists and radiation dosages of each.

Starship assumptions

The mission profile uses a modified SpaceX Starship Block 3 as the primary Europa landing spacecraft, along with two Starship depots, and the Starship tanker, both under development for the Artemis program[12]. As the Europa lander will not enter an atmosphere post launch, it does not need TPS (a thermal protection system), nor aerosurfaces present on normal Starships. We take the conservative performance estimate of the performance of the Europa lander to have a propellant load of 1600t[13], a dry mass of 110T and a payload of 10t, with raptor vacuum engines with a specific impulse of 370-80s. The 10t dry mass reduction compared with a normal Starship is due to the removal of the TPS. We take the dry mass of the Europa lander at 110t as a conservative estimate considering the first orbital prototype had a dry mass of 100t[14], with the caveat that this was a block 1 ship with a less voluminous tank section. A ground up estimate places the dry mass of a block 2 ship at 126t[15]. The removal of TPS and aero surfaces is almost certainly significant enough to drop the mass by 16t. The depot Starships and tanker Starship are assumed to have the same propellant load and each LEO launch is assumed to leave 100T of extra propellant in the tanker upon arrival in LEO (Low Earth Orbit). The Earth ejection refueling profile takes the conservative performance estimates. We also take optimistic performance estimates, with a Europa lander dry mass of 85T, raptor vacuum specific impulse of 380s, but the same propellant mass.

Trajectory methods

The Earth-Jupiter trajectories were calculated with patched conics 2-body approximations using the poliastro python library[16][17]. As is standard in preliminary Jupiter mission architecture studies[18][19][20], we employ Lambert-arc solutions for Earth-Jupiter and moon to moon transfers, effectively treating the spacecraft's heliocentric leg as a two-body problem (Sun/Jupiter + vehicle), and each hyperbolic planet exit/entry as a two-body problem (Earth/Jupiter + vehicle). Each Galilean Moon gravity assist is modeled as solely bending the V_∞ vector by a given pump angle, with the crank angle set to approximately zero (essentially restricting the flybys to the ecliptic plane). The positions of the celestial bodies were obtained from JPL ephems python library[21][22]. Higher-order multi-body perturbations are negligible for initial Earth-Jupiter DV estimates. Higher order perturbations are not negligible for the Jovian moon-moon transfers, but initial patched conics analyzes with subsequent fine graining with higher order models remain consistent with the approach used historically by NASA. We

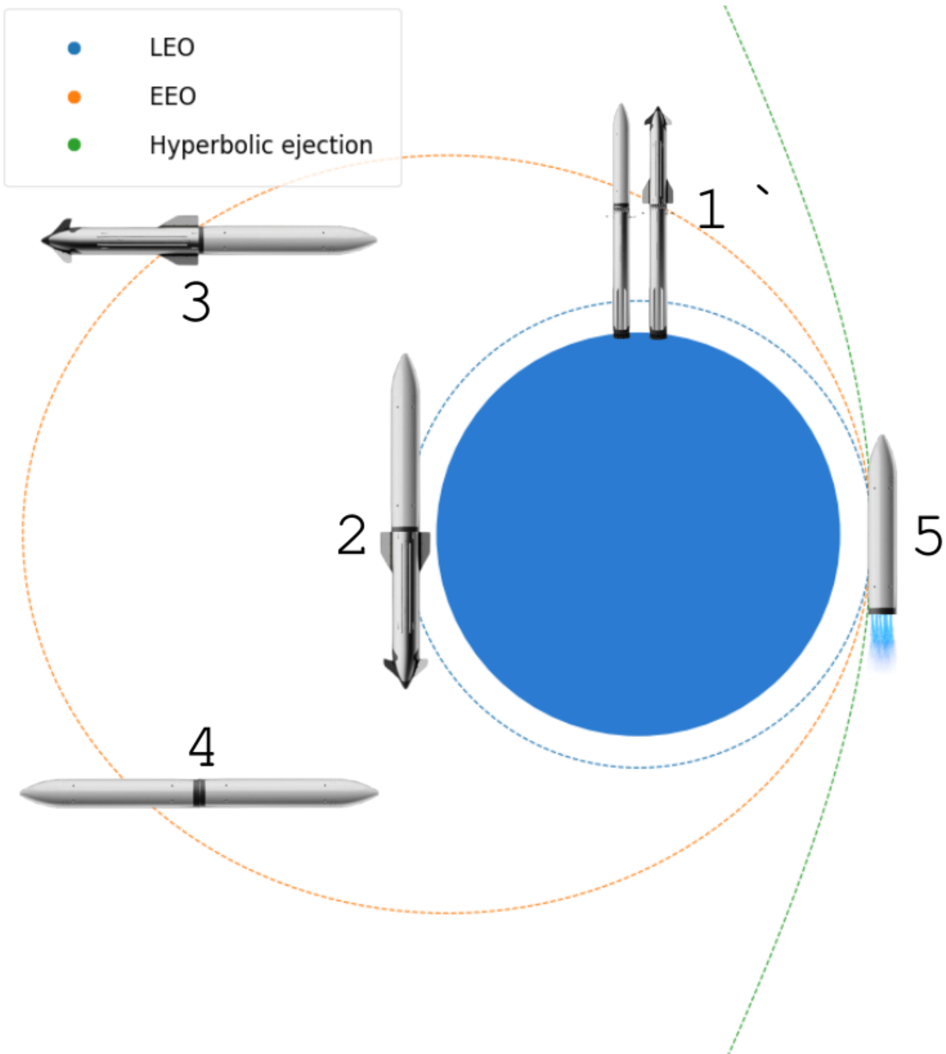

FIGURE 1.

iterated the poliastro Lambert-arc solver over 1-day timesteps in the 2030s to look for the lowest Earth-Jupiter ejection DV solutions in the Sun reference frame.

Earth departure

The Earth departure involves 5 steps, shown in Fig 1:

(1) Launch from Earth of two depot ships and 37 tanker refuels.
(2) Refueling of a depot ship in a 300km LEO (Low Earth Orbit).
(3) Transfer of LEO depot's fuel via a tanker to the EEO (Elliptical Earth Orbit) depot.
(4) Launch and refueling of Europa lander via EEO depot.
(5) Europa lander burns for Jupiter.

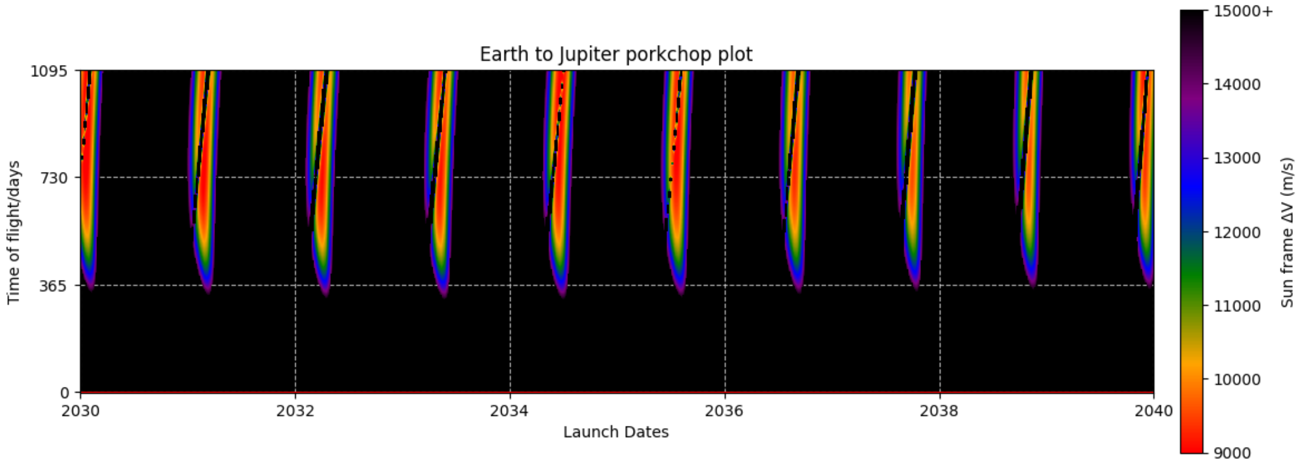

Figure 2.

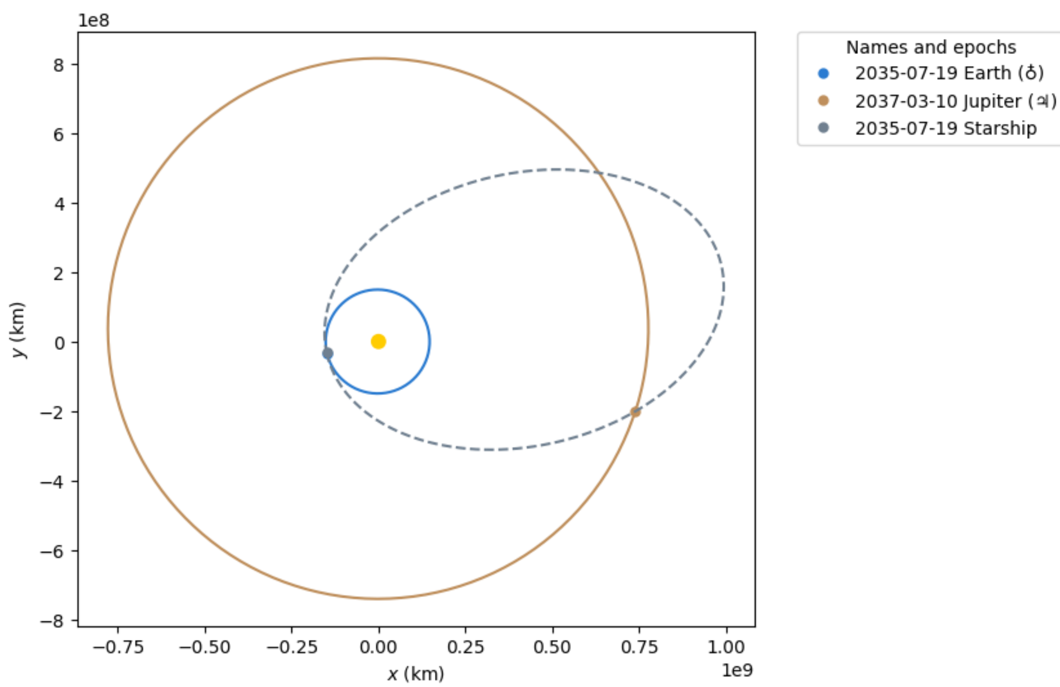

Figure 3. Earth-Jupiter transit

The transfer to Jupiter takes 600 days in the prime trajectory departing on the 19th July 2035, but launch windows exist with similar energy requirements every $\sim$400 days. These launch windows are plotted in the Porkchop plot in Fig 2. The baseline Earth-Jupiter trajectory is plotted in Fig 3.

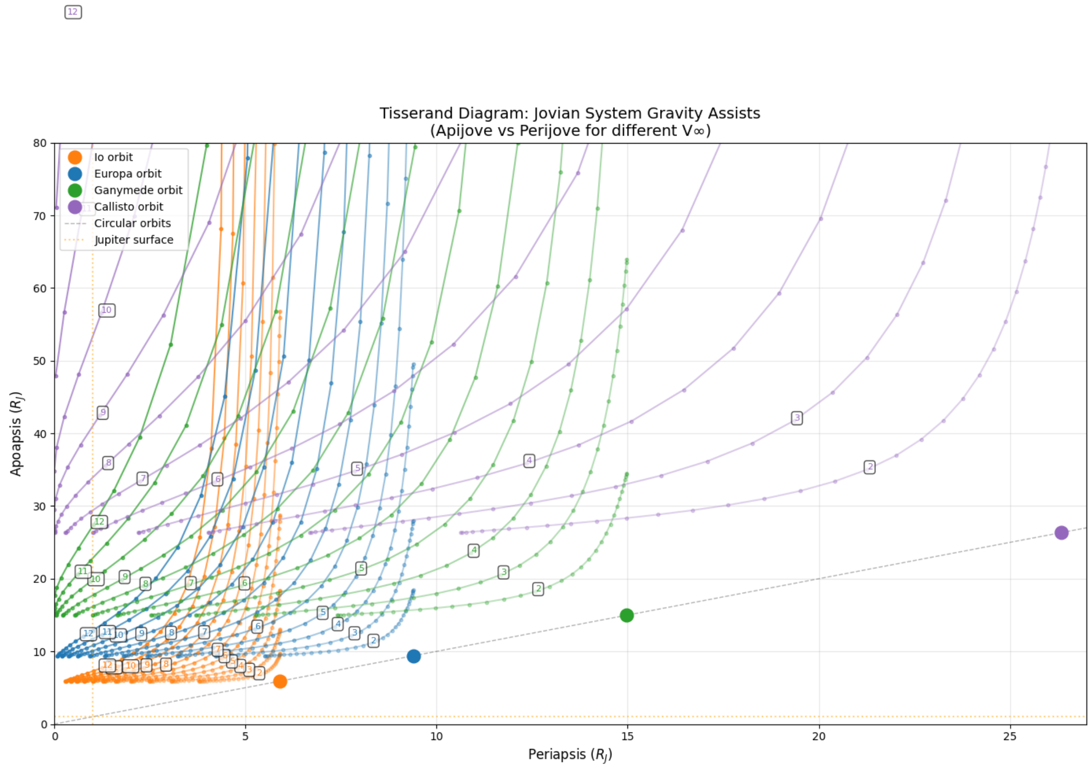

FIGURE 4. Jovian Tisserand plot. The numbered contours represent possible trajectories with constant V_∞ relative to the given Galilean satellites.

Jupiter arrival

We baseline an arrival perijove of $1.05R_j$ (1.05 Jovian radii) allowing for effective use of the Oberth Effect. This is approximately the same initial perijove that the Juno spacecraft used during its orbital insertion in 2016[23]. An approximately 1km/s DV Jupiter orbit insertion (JOI) burn occurs at first perijove. The fast, baseline and slow trajectories diverge at Jupiter arrival, trading DV for time of flight and or reduced radiation dosage. Because, even with conservative dry mass assumptions, Starship has a large amount of DV (vastly in excess of any historical Jovian probe[24][25][26]), the initial capture orbit can be relatively low period compared to historical capture orbits. After an apijove raising maneuver, the assist period begins.

Tisserand plots and Jovian tour design

Fig 4 shows the energy contours of possible trajectories that fly past the Galilean moons. Due to the conservation of energy, the spacecraft must leave the moon's sphere of influence with as much velocity ($|V_\infty|$) relative to the moon as it had coming in. The magnitude, $|V_\infty|$, is unchanged by each flyby. It will however be rotated by an angle[27]:

$$\theta = 2sin^{-1}\left(\frac{1}{V_\infty^2 r_p/\mu + 1}\right)$$

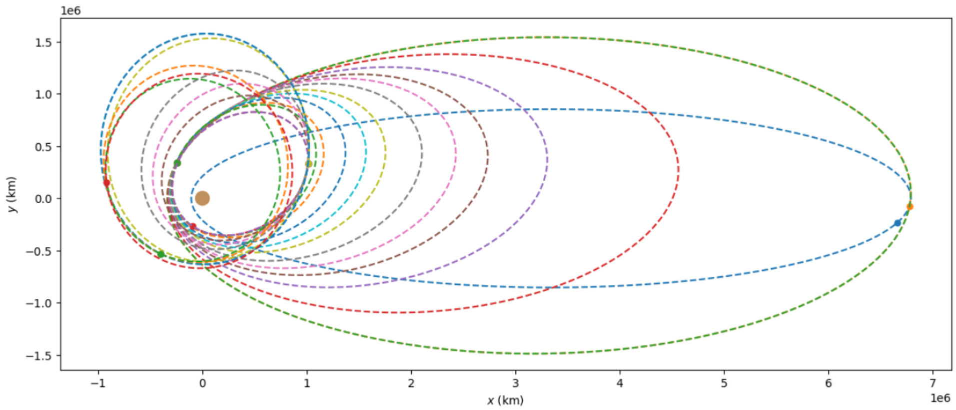

FIGURE 5. Europa lander's orbits as the gravitational assists deflect the orbit to the final Europa encounter.

Thus we can plot a set of orbits, each with the same $|V_\infty|$ relative to a moon at flyby as a contour of orbits with different perijoves and apijoves. Each assist rotates the V_∞ vector to a new orbit along that contour. From this we can plot a path that brings the spacecraft to Europa with a low enough V_∞ for an economical orbital insertion. We baseline 10 gravity assists with Io reducing the apijove, four gravity assists with Ganymede raising the perijove, two further assists with Europa, and one final assist with Ganymede, arriving at Europa 9 months after JOI, and 2.5 years after launch. The baseline Jovian tour orbits are plotted in Fig 5. Assuming 10t of payload, and 380s ISP Raptor Vacuum engines, the dry mass of the ship must be less than 85t, 113t and 150t to be capable of achieving the rapid, baseline and slow trajectories respectively. With 10t of payload but 370s Raptor Vacuum ISP, this is 78t, 105t and 140t.

Integration of the mission radiation dose

Jupiter's extremely strong magnetic field constrains high energy charged particles into orbits close to the planet. The flux of especially high energy electrons can seriously damage electronics, notably damaging the Galileo spacecraft[31] with radiation constraining star trackers, damaging MOSFET gates connected to the gyros, causing local ESD discharge on sliprings and slowly degrading spacecraft systems. We used the SPENVIS[32] D&G 1983 Jovian radiation model to integrate the electron, proton flux and absorbed radiation dose during the Jovian mission. The Starship hull is composed of 4mm thick stainless steel[33]. The SPENVIS model predicts that electronics (silicon) behind the iron (SPENVIS does not model steel) take approx $\sim 20krad$ during the initial capture orbit. Io flybys take $\sim 10krad$, and this dosage is reduced further out. The total radiation dosage comes out to $\sim 200krad$ in the baseline trajectory, $\sim 60krad$ in the rapid trajectory and $\sim 100krad$ in the slower trajectory. This is less than the Galileo spacecraft[31] took over its lifetime ($\sim 600krad$). While we appreciate that these are simplified calculations, approximating the Starship hull as a 4mm thick flat shield of iron, we believe they show radiation dosage/electronics survivability is a manageable problem, within precedent for historical Jovian missions.

Cryogenic propellant boiloff

The cryogenic LOX and LCH_4 propellant required for outer planet insertion and landing maneuvers must remain storable during interplanetary cruise. For Earth–Jupiter missions, it is sufficient to show that the thermal load is manageable near Earth, since the solar irradiance decreases with distance to the Sun and becomes less challenging en route to Jupiter.

We consider a Starship-derived vehicle without TPS or aerosurfaces, with an exterior coated in a high-emittance spacecraft white thermal-control paint. A representative coating is Z-93P, with solar absorptance $\alpha = 0.136$ and thermal emittance $\epsilon = 0.921$ [28]

In the thermally favorable cruise attitude, the vehicle points toward the Sun, so the solar-illuminated projected area is approximately the circular cross-section

$$A_{\mathrm{proj}} = \pi \left(\frac{D}{2}\right)^2$$

Taking $D = 9$ m gives $A_{\mathrm{proj}} = \pi(4.5 \text{ m})^2 = 63.6 \text{ m}^2$. The absorbed solar power at 1 AU is then

$$P_{\mathrm{abs}} = \alpha S_0 A_{\mathrm{proj}}$$

with $S_0 = 1361 \text{ W m}^{-2}$, yielding

$$P_{\mathrm{abs}} = 0.136 \cdot 1361 \cdot 63.6 = 1.18 \cdot 10^4 \text{ W}$$

Crucially, this absorbed power does not need to be rejected by dedicated radiators if it can be confined to the Sun-facing forward section (nose/payload compartment) and radiated directly to space. The radiative emission available from a painted surface at temperature T is

$$P_{\mathrm{emit}} = \epsilon \sigma A T^4$$

so the radiating area required for the forward section to re-radiate the absorbed load while remaining at a benign $T \sim 300$ K is

$$A_{\mathrm{req}} = \frac{P_{\mathrm{abs}}}{\epsilon \sigma T^4}$$

Using $\epsilon = 0.921$ and $T = 300$ K gives $\epsilon \sigma T^4 \approx 4.23 \cdot 10^2 \text{ W m}^{-2}$, hence

$$A_{\mathrm{req}} \approx \frac{1.18 \cdot 10^4}{4.23 \cdot 10^2} \approx 2.8 \cdot 10^1 \text{ m}^2$$

Therefore only $\sim 28 \text{ m}^2$ of effective radiating area at ~ 300 K is required to passively reject the Earth heliocentric distance solar load. The available external area of the nose/forward section is naturally far larger than this, implying that maintaining the forward section near room temperature is feasible without imposing kW heat flow into the cryogenic tanks.

To suppress radiative coupling from the warm forward section into the tank section, we adopt the same simplifying isolation argument we used for inner-solar-system transits[29]: a low-emissivity polished aluminum thermal break plate between the forward section and the tanks. For a polished aluminum surface with emissivity $\epsilon_{\mathrm{Al}} \sim 0.06$, the radiative power emitted toward the tank section (bounded by the cross-sectional area) at $T \sim 300$ K is

$$P_{\mathrm{Al} \to \mathrm{tanks}} \sim \epsilon_{\mathrm{Al}} \sigma T^4 A_{\mathrm{proj}} \approx 0.06 \cdot \sigma \cdot 300^4 \cdot 63.6 \approx 1.7 \cdot 10^3 \text{ W}$$

which is small compared to the tank-section radiative rejection capacity.

Conduction through the steel skin from the forward section into the cryogenic tank barrel is also negligible. Using a stainless thermal conductivity of $\sim 15 \text{ W K}^{-1} \text{ m}^{-1}$ near room temperature [30], which decreases at cryogenic temperature, and taking the temperature drop to occur over the length of one ring segment, the conducted power is $\sim 400W$, smaller than the radiative leakage estimated above.

A simple bound on the passive heat rejection of the cryogenic tank section is the radiative emission if the external tank surfaces are at the LOX saturation temperature. Taking $T_o =$ 113 K and assuming the exterior is coated in Z-93P the emitted power from the combined tank barrel area A_t is

$$P_{\text{tanks}} = \epsilon\sigma A_t T_o^4$$

Using the same cylindrical area approximation as above ($A_t = A_m + A_o \simeq 310 + 620 = 930 \text{ m}^2$) gives an emitted heat flux

$$\epsilon\sigma T_o^4 = 0.921\sigma(113 \text{ K})^4 \approx 8.52 \text{ W m}^{-2}$$

and therefore

$$P_{\text{tanks}} \approx 8.52 \text{ W m}^{-2} \cdot 930 \text{ m}^2 \approx 7.9 \cdot 10^3 \text{ W}$$

Thus, at 113 K the tank section radiates $\sim$ 8 kW to deep space,

In summary, near Earth the Sun-pointing attitude limits absorbed power to $\sim$ 12 kW, which can be re-radiated directly by a white-painted forward section at $\sim$ 300 K using only $\sim$ 28 m^2 of effective radiating area. With a low-emissivity polished aluminum radiation break and the low conductive coupling of the steel structure, the thermal load can be confined to the forward section rather than driving substantial heat input to the cryogenic tanks. Since solar irradiance decreases as $1/r^2$, the thermal environment becomes less demanding toward Jupiter, and satisfying the near-Earth bounding case is sufficient for the solar-heating component of the cryogenic storage problem. We acknowledge that these calculations are simplified, modeling the Starship as a cylinder and ignoring the challenges of nose cone thermal management, but we believe they show Starship interplanetary transit cryogenic storage is a manageable problem.

Europa Landing

The Starship arrives with a V_∞ of 1.8km/s at Europa, and performs a 1.3km/s Europa Orbit Insertion burn to enter a 50km circular orbit of Europa where final pre landing checks can be made. With the baseline Jovian tour trajectory and the assumptions about Starship performance, 70T of propellant remains in the tanks. We assume a landing burn with 1 active sea level (SL) raptor engine. This engine has a specific impulse of 350s[34]. The three raptor vacuum engines do not have gimbal control, and we assume that this is necessary for the landing burn. It is possible that sufficient control authority could be obtained through differential throttling of the vacuum optimized engines, but as sufficient margin exists to land with the less efficient sea level engine, we baseline it. The SL raptor engine can deep throttle to atleast 50% of its rated thrust - 140T[35][36]. At the start of the burn, the ship will have a minimum thrust to weight (on Europa) ratio of 4.8, increasing as the propellant tanks empty. The inability to provide less thrust than the gravitational force on the lander forces the lander to adopt a hoverslam trajectory. We consider a small descent orbit insertion burn that places the spacecraft on a trajectory with a periapsis just below Europa's surface. Once the hoverslam threshold is crossed, the sea level engine ignites, burning continuously until touchdown. This type of high thrust to weight ratio landing burn is required on the Falcon 9, and is thus not without precedent. We calculate that in the baseline case, the landing uses 1.47km/s of DV and 66T of propellant, with 4T reserve. A caveat worth stating is that raptor plume impingement is still a major engineering unknown, even if the surface is mechanically "cleaner" than loose regolith on the Moon or Mars. Europa's solid ice may reduce dust-driven erosion and particulate ejecta, but understanding the degree of excavation will require detailed modeling.

Implications

By demonstrating that Starship can carry heavy payloads to the surface of Europa, substantial planning for Starship sized surface payloads can begin. 10T is in excess of any payload ever delivered to the surface of any celestial body (other than Earth) in history - even the mass of the landed Apollo LM[37] was only 8.2T. If Starship demonstrates the required capabilities; orbital refueling, radiation hardening, bioburden elimination, long distance communication systems and low enough dry mass, the implications for exploration of Europa will be monumental. Compared to historical plans - see table 1, planetary scientists will be free to try audacious plans, trading mass to overcome engineering challenges.

Name	**Time to reach Europa**	**Surface science payload description**	**Science payload (kg)**
Europa Clipper (NASA orbiter)	$\approx$ 5.5 yr to Jupiter system	None (Flybys; remote sensing of ice/ocean)	0
JUICE (ESA orbiter)	$\approx$ 8.3 yr to Jupiter system	None (Flybys; remote sensing)	0
Europa Lander (NASA flagship concept; pre-Phase A)	$\sim$ 6.5 yr [4]	Cameras; microscopy; spectroscopy and mass spectrometry for chemistry; sample acquisition	42.5
Starship (baseline trajectory)	2.5 yr	Massive science equipment; possibly cryobots, submarines	$\sim$ 10000

Table 1. Representative Europa mission plans and concepts

Launch cadence

Because the trajectory demands dozens of refuellings, high launch cadence must be achieved. The $\tilde{4}0$ launches is just under one quarter of SpaceX's 2025 flight rate, implying that if cadence did not improve beyond 2025 levels, the refuelling campaign would take on the order of three months. This obviously ignores operational constraints novel to Starship, including pad availability, range scheduling and reliability issues which could lengthen the refuel campaign. Accordingly, the three-month figure should be treated as an order-of-magnitude estimate based on a realistic Starship launch cadence, noting that demand from programs such as Artemis and Starlink could also push cadence higher.

Planetary protection concerns

The primary goal for Europa exploration is the search for extra-terrestrial life. It would be catastrophic if the pursuit of this led to contamination of the ocean with terrestrial life that either conflicted with the Europan life and/or made distinguishment of the Europan life impossible. NASA has set the rule that missions to Europa must have less than a 1/10000 chance of contaminating the ocean [38]. As sterilizing the instruments designed to contact the ice is well within robotic mission precedent, the primary contamination risk for a surface exploration only mission is the vehicle crashing into Europa during a failed landing. The probability of contaminating the ocean after contamination of the surface of Europa is unknown and dependent on mechanisms that transfer material from the surface into the ocean. While we hope that the Europa Clipper mission provides more insight into this, assuming that said processes do exist, to achieve sufficient vehicle sterilization:

(1) The entire interior and exterior must be totally biologically sealed from eachother, such that when the vehicle is brought outside and stacked for launch, the sterilized interior remains as such.
(2) The interior must be sufficiently sterilized, including pressurized and radiation shielded areas.
(3) The exterior must be sufficiently sterilized by 2 years of exposure to vacuum and significant radiation exposure upon Jovian arrival. The exterior will receive an excess of 10Mrad and as in NASA's 2016 Europa lander mission plan[4], we consider this to have achieved the required microbial cleanliness.

We recognize that these are significant engineering challenges, especially for Starship, which was not originally designed to be totally free of bio-contaminants. We believe the massive benefits of Starship for a mission to Europa significantly outweigh the engineering challenges brought to achieve planetary protection.

Conclusion

This work argues that Europa exploration has been primarily constrained by the propulsive energy required to reach Europa, forcing traditional mission architectures into extreme mass austerity that massively constrains science ambition. Our results suggest that the SpaceX Starship, with reasonable assumptions about performance, has the propulsive energy to land much heavier payloads on the surface of Europa on fast (1-4 yr) Earth to Europa trajectories. The multi tonne payload allows for heavy radiation shielding, larger communication systems and dramatically expanded experimental ambitions.

Nevertheless several considerations highlight that this approach, while promising, is not free of challenges. The high launch cadence and propellant transfer demands required for dozens of orbital refueling flights introduce significant operational risk and assume a level of orbital logistics maturity that has not yet been demonstrated. Second, Jovian radiation remains a major constraint: shielding mass is available, but system survivability depends on detailed trajectory dependent dose, electronics qualification, and fault tolerant architectures.

Third, planetary protection compliance for Starship is a substantial engineering challenge. Crushing bioburden on a large, complex vehicle with hundreds of welds, joints, feedlines, valves and sheltered cavities is fundamentally harder than for a compact landers built around high-temperature sterilization (eg Viking). Systems may not be compatible with blanket heat sterilization, and there is significant recontamination risk during integration, fueling, and launch operations. Substantially more engineering work is required to develop an effective bioburden control approach before sending a Starship to Europa. We therefore view this study as an initial framework outlining the key architectural elements required to send heavy payloads to Europa, recognizing that higher fidelity analyzes are necessary to fully validate the concept. With these caveats in mind, the central implication remains: Starship plausibly moves Europa surface exploration out of extreme mass austerity and towards ambitious surface science. If the operational and planetary protection constraints can be closed, Starship's capability may bring forth Europa surface missions where science return is limited less by kilograms and more by how effectively we choose to spend them.

Funding

The author did not receive funding for this research project.

Appendix A - Trajectory

	Baseline (380 s Isp)	Fast (380 s Isp)	Slow (370 s Isp)
Total DV (km/s)	9.8	10.8	8.8
DV used pre-landing (km/s)	8.3	9.3	7.3
Propellant pre-landing (t)	70.5	53.9	110.0
DV free for landing (km/s)	1.5	1.7	2.1
Landing DV (350 s Isp) (km/s)	1.5	1.5	1.5
Starship dry mass (t)	110.0	80.0	120.0
Payload (t)	10.0	5.0	10.0
Starship propellant load (t)	1600.0	1600.0	1600.0
4 mm steel radiation dose total (krad)	220	65	103
TJI $\rightarrow$ JOI time (d)	600	600	750
JOI $\rightarrow$ EOI time (d)	308	164	727
Total time to Europa (d)	908	764	1477

TABLE 2. Delta-V, mass, timing, and radiation summary for candidate Europa trajectories

Event	Type	Δv (m/s)	Periapsis (km)	Date
Time to reach Europa	—	—	—	908 d
Total Δv	—	9797	—	—
TJI	Burn	4103	—	19-07-35
JOI Burn	Burn	1000	—	10-03-37
Perijove Raise Burn	Burn	688	—	04-04-37
Flyby 1 (Io)	Gravity Assist	—	275.52	25-04-37
Flyby 2 (Io)	Gravity Assist	—	369.35	19-05-37
Flyby 3 (Io)	Gravity Assist	—	1256.75	04-06-37
Flyby 4 (Io)	Gravity Assist	—	2232.49	17-06-37
Flyby 5 (Io)	Gravity Assist	—	1249.76	27-06-37
Flyby 6 (Io)	Gravity Assist	—	372.85	06-07-37
Flyby 7 (Io)	Gravity Assist	—	1403.49	13-07-37
Flyby 8 (Io)	Gravity Assist	—	754.16	26-07-37
Flyby 9 (Io)	Gravity Assist	—	128.79	31-07-37
Flyby 10 (Io)	Gravity Assist	—	3007.59	09-08-37
Ganymede encounter burn	Burn	281	—	13-10-37
Flyby 11 (Ganymede)	Gravity Assist	—	118.30	16-10-37
Flyby 12 (Ganymede)	Gravity Assist	—	3203.24	07-11-37
Flyby 13 (Ganymede)	Gravity Assist	—	3604.02	21-11-37
Flyby 14 (Ganymede)	Gravity Assist	—	357.87	12-12-37
Flyby 15 (Ganymede)	Gravity Assist	—	4000	19-12-37
Europa encounter burn	Burn	575	—	28-12-37
Europa orbit insertion	Burn	1650	—	12-01-38
Europa landing	Burn	1500	—	—

TABLE 3. Baseline mission timeline and key burns / flybys

Event	Type	Δv (m/s)	Periapsis (km)	Date
Time to reach Europa	—	—	—	764 d
Radiation dosage	—	—	—	65 krad
Total Δv	—	10778	—	—
TJI	Burn	4103	—	19-07-35
JOI Burn	Burn	900	—	10-03-37
Perijove Raise Burn	Burn	861	—	04-04-37
Flyby 1 (Ganymede)	Gravity Assist	—	29.96	29-05-37
Flyby 2 (Ganymede)	Gravity Assist	—	—	03-07-37
Flyby 3 (Ganymede)	Gravity Assist	—	—	25-07-37
Flyby 4 (Ganymede)	Gravity Assist	—	—	08-08-37
Flyby 5 (Ganymede)	Gravity Assist	—	—	30-08-37
Europa encounter burn	Burn	354	—	12-08-37
Europa orbit insertion	Burn	3060	—	21-08-37
Europa landing	Burn	1500	—	—

TABLE 4. Fast mission timeline and key burns / flybys

Event	Type	Δv (m/s)	Periapsis (km)	Date
Time to reach Europa	—	—	—	1477 d
Radiation dosage	—	—	—	103 krad
Total Δv	—	8781	—	—
Δv free for landing	—	7281	—	—
TJI	Burn	3745	—	18-07-35
JOI Burn	Burn	550	—	10-03-37
Perijove Raise Burn	Burn	622	—	04-04-37
Flyby 1 (Ganymede)	Gravity Assist	—	75.88	06-07-37
Flyby 2 (Ganymede)	Gravity Assist	—	1204.84	14-07-37
Flyby 3 (Ganymede)	Gravity Assist	—	1489.83	11-08-37
Flyby 4 (Ganymede)	Gravity Assist	—	3058.00	02-09-37
Flyby 5 (Ganymede)	Gravity Assist	—	1343.10	08-10-37
Flyby 6 (Ganymede)	Gravity Assist	—	50.00	22-10-37
Callisto encounter burn	Burn	139	—	26-11-37
Flyby 7 (Callisto)	Gravity Assist	—	4308.75	15-01-38
Flyby 8 (Callisto)	Gravity Assist	—	2887.31	09-04-38
Flyby 9 (Callisto)	Gravity Assist	—	1052.12	25-04-38
Flyby 10 (Callisto)	Gravity Assist	—	240.09	—
Europa encounter burn	Burn	575	—	—
Europa orbit insertion	Burn	1650	—	12-01-38
Europa landing	Burn	1500	—	—

TABLE 5. Slow mission timeline and key burns / flybys

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